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Objective of AOA :-

The objective of AOA is to evaluate algorithm performance and efficiency in terms of time and space complexity.

An efficient algorithm is the ~~one~~ one with lesser time and space complexity.

The analysis helps in determining the amount of resources required to execute an algorithm, which is crucial for selecting the most efficient algorithm for a specific problem.

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Scope of Analysis of algorithm :-

It covers understanding, evaluating and comparing algorithms to determine their efficiency, scalability and suitability for solving computational problems.

1 → PURPOSE

* measure performance in terms of :-
  (a) Time complexity (execution time growth with input size)
  (b) Space complexity (memory usage)

* predict scalability before implementation
* compare algorithms for the same problem to choose the most optimal one.

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2. Areas covered

(a) performance analysis :-
→ worst case, average case, best-case analysis
→ asymtotic notations :- Big-O, Big Ω, Big Θ

(b) ~~CPU~~ Resource utilization
→ CPU time, memory, disk usage, n/w bandwidth

(c) Algorithm classification
→ Recursive v/s iterative
→ Deterministic v/s non-deterministic
→ Exact v/s approximate

(d) Trade-off analysis
→ Time v/s space trade offs
→ pre-processing v/s query time

(e) Scalability
→ how performance changes ~ large i/ps

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3) Applications

→ S/w development :- choosing efficient algo for performance critical systems.

→ data structures :- selecting optimal structures for storage and retrieval.

→ competitive programming :- writing code that runs within strict time limits.

→ big data and AI :- handling massive datasets efficiently

→ embedded systems :- optimizing for limited memory and processing power

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4) Benefits

→ Reduces execution and resource costs
→ Improves user experience in applications
→ Ensures feasibility for large-scale problems
→ helps in theoritical research and innovation in computing

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Outcome of AOA

→ understanding time and space complexity of algorithms, which helps in evaluating their efficiency.

→ Learning about various data structures, s.a arrays, linked lists, stacks, queues, trees and graphs along with their applications.

→ gaining knowledge on searching and sorting algorithms, which are fundamental for efficient data manipulation.

→ familiarity with algorithm design paradigms and techniques for analyzing algorithms including lower and upper bounds

→ practical implementations to solidify understanding and prepare for technical interviews

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These outcomes are essential for developing a strong foundation in algorithm analysis and design.

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